

Qianqian Tong

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EDUCATION

University of Texas at Austin

Ph.D. in Civil Engineering

Austin, TX

Aug 2023 – May 2027 (expected)

Shanghai University of Engineering Science

B.E. in Transportation Engineering, outstanding undergraduate student

Shanghai, China

Sep 2019 – Jul 2023

WORK EXPERIENCE

ByteDance

Jun 2023 – Aug 2023

Algorithm Engineer Intern (Multimodal)

Shanghai, China

- Worked on multimodal representation learning within short-video content understanding pipelines, supporting feature generation for downstream recommendation and search systems
- Integrated Segment Anything Model (SAM) into offline video and image preprocessing pipelines, redesigning visual feature extraction by replacing object-detection-based features with region-level segmentation outputs
- Assessed the feasibility of segmentation-based visual features replacing detection-based features in large-scale preprocessing pipelines by comparing downstream representation quality and per-video preprocessing cost

Ping An Technology

Jan 2023 – Mar 2023

Algorithm Engineer Intern (Research)

Shanghai, China

- Designed a multimodal image captioning training framework that integrates image features, predicted semantic labels, and label probability vectors to improve feature importance modeling and gradient-based optimization
- Implemented a differentiable feature-label fusion mechanism enabling end-to-end backpropagation in generative language models, leading to improved image-text alignment and caption accuracy over baseline training methods
- Extended the framework with ViT architectures to enhance representation learning and generalization across application scenarios, and contributed to a granted patent on image description sentence generation systems

RESEARCH AND PROJECTS

University of Texas at Austin

Aug 2023 – Present

Research Assistant, Multimodal Transportation Systems

Austin, TX

- Designed a network-based discrete-event simulation framework to model statewide multimodal freight operations across marine, rail, and roadway systems, formulating terminal layout and congestion modeling assumptions to minimize total vehicle travel distance and capture operational dynamics, supporting planning-level analysis for the Parallel Systems autonomous, battery-electric rail system
- Developed multimodal terminal queuing models using MDP formulations to characterize state transitions and decision dynamics, enabling policy optimization for commodity destination splitting under cost and delay objectives
- Implemented freight assignment and routing models to evaluate system performance under user-equilibrium conditions at the statewide level and shortest-path routing for nationwide network applications

National Laboratory of the Rockies (NREL)

Sep 2023 – Jul 2026

Research Assistant, Intermodal Freight Optimization (INFORMES)

Golden, CO

- Developed a SimPy-based intermodal terminal simulation framework (LIFTS) to model container flows, resource allocation, and event-driven operations across terminal nodes, corridors, and networks
- Built data pipelines to generate and visualize track chart attributes (grade, curvature, speed limits, siding locations) and encode into standardized YAML topology inputs for consistent ALTRIOS corridor simulations
- Integrated terminal-level simulations with nationwide rail network models within the ALTRIOS framework to evaluate system-wide performance output metrics, including container processing time distributions, delay propagation, operational costs, and energy consumptions

Tongji University

Jul 2021 – Aug 2022

Research Assistant, Machine Learning

Shanghai, China

- Designed a two-stage tumor subregion modeling framework using voxel-level FDG PET/CT data to identify high-risk and low-risk regions for chemoradiation response assessment in non-small cell lung cancer
- Implemented AutoML-based classification and regression models with conventional and radiomics features, achieving RMSE to 0.17 under leave-one-patient-out cross-validation
- Optimized clustering and prediction performance via feature weighting and Bayesian optimization, improving silhouette coefficient by 8–10% and outperforming conventional machine learning methods on 25 patient datasets

Rail Transit Testing Technology Laboratory

Apr 2020 – May 2022

Research Assistant, Computer Vision

Shanghai, China

- Implemented image preprocessing and region extraction pipelines using HALCON and Python
- Built an end-to-end computer vision system for railway track inspection using YOLOv3 and BASNet
- Developed a PyQt-based visualization interface and designed a camera-mounted railway inspection prototype

TECHNICAL SKILLS

Programming: Python (PyTorch, JAX, SciPy, Polars, SimPy, NumPy, etc), C, JavaScript, R, Rust, SQL

Skills: Computer Vision (Vision Transformers, multimodal representation learning, object detection, CNNs), Scientific Machine Learning (physics-informed learning, simulation-driven modeling), Reinforcement Learning (MDP formulation, policy optimization, simulation-based training)